

Real-Time Impact Point Prediction of Suborbital Rockets Using Kalman Filtering

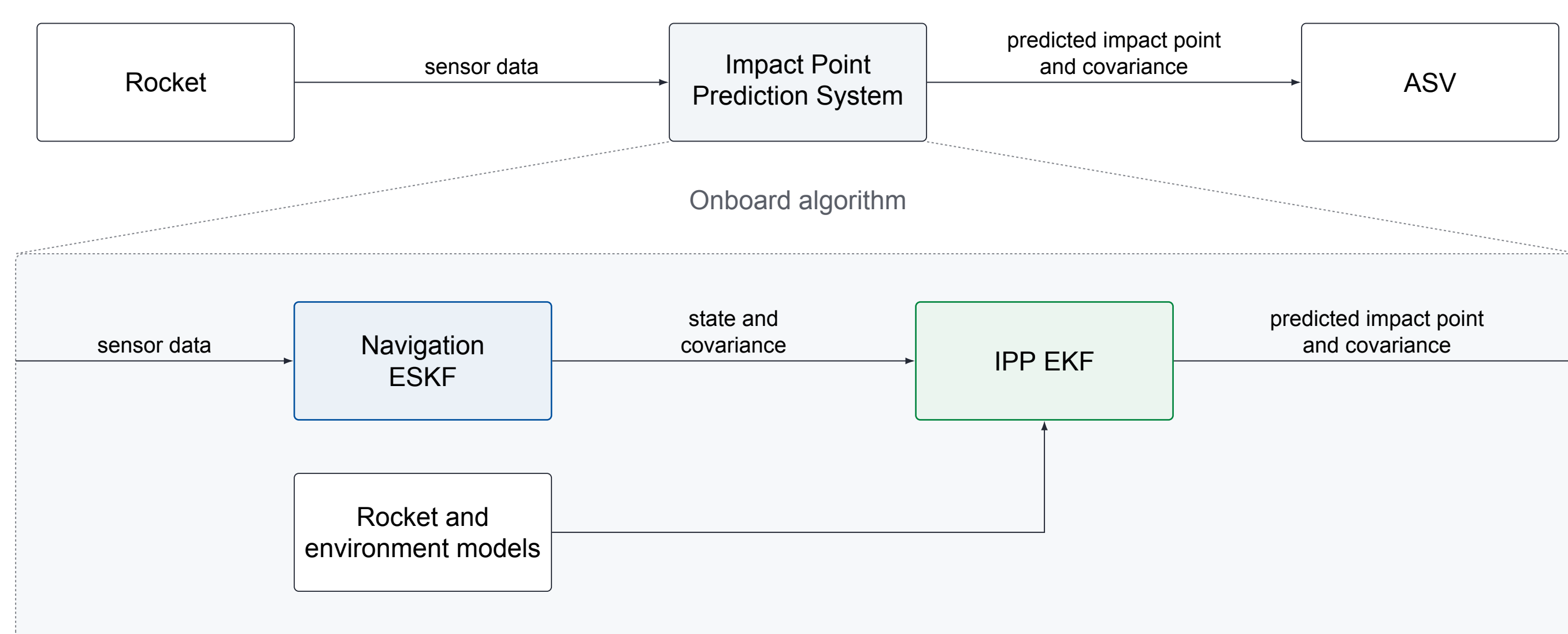
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Research Objective

Rocket launch rates are rising, making sustainable recovery of payloads and first stages increasingly important. Autonomous surface vehicles (ASVs) can reduce cost, human exposure, and environmental impact, but require real-time prediction of the rocket landing location. This work focuses on impact point prediction (IPP), which continuously estimates the landing location and uncertainty from onboard sensor data. The ASV uses this estimate for recovery guidance.

IPP Architecture



Navigation filter. The Error-State Kalman Filter (ESKF) estimates position, velocity, attitude, and sensor biases from inertial propagation with GNSS, barometer, and magnetometer aiding. Its state and covariance initialise the IPP filter.

Impact point prediction filter. The IPP filter is an Extended Kalman Filter (EKF). It propagates a Cartesian state $\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T]^T$ in NED coordinates forward until altitude reaches sea level:

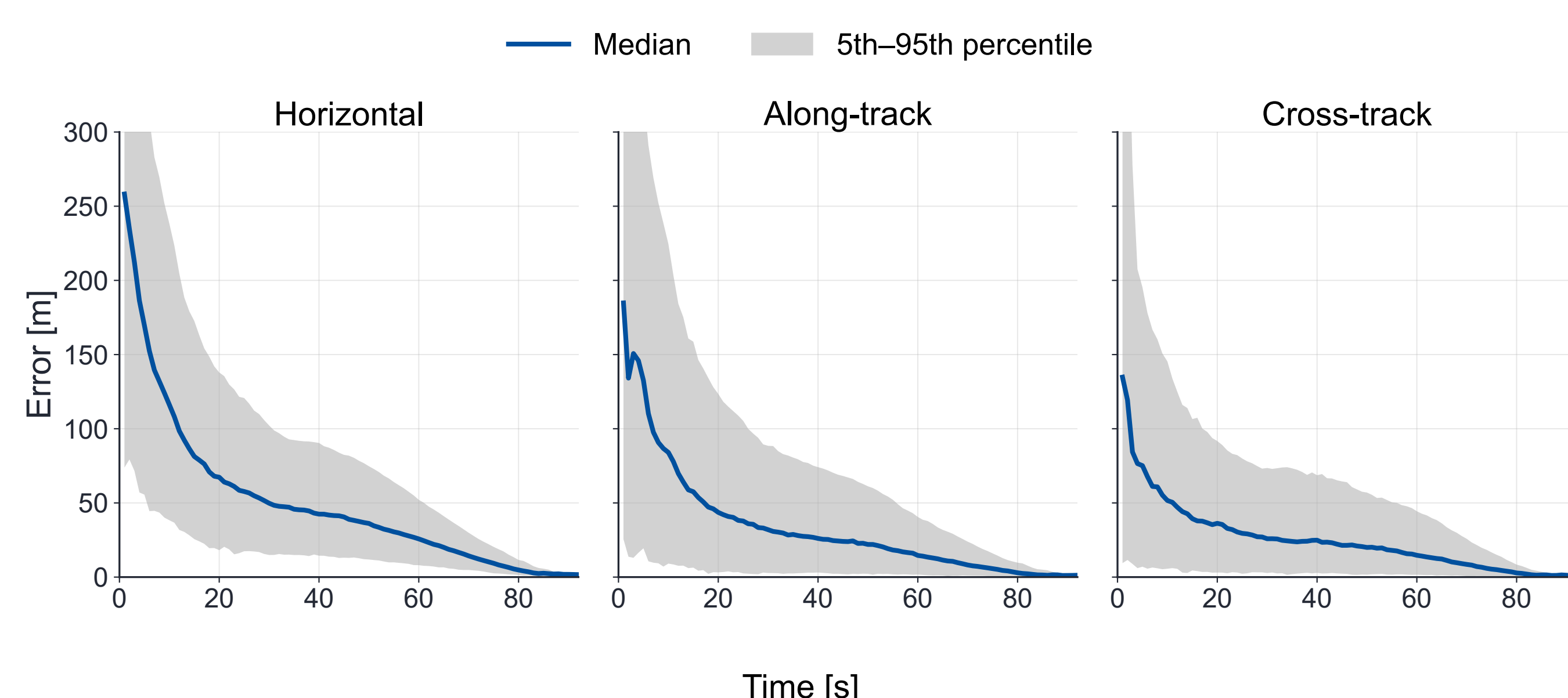
$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{g} - K_D \|\mathbf{v}_a\| \mathbf{v}_a + \mathbf{R}(\hat{\mathbf{q}}) \mathbf{a}_T, \quad K_D = \frac{\rho C_D A}{2m}$$

The acceleration model includes gravity, drag, and thrust. The body-frame thrust $\mathbf{a}_T$ is rotated to NED by $\mathbf{R}(\hat{\mathbf{q}})$, where $\hat{\mathbf{q}}$ is the ESKF attitude quaternion.

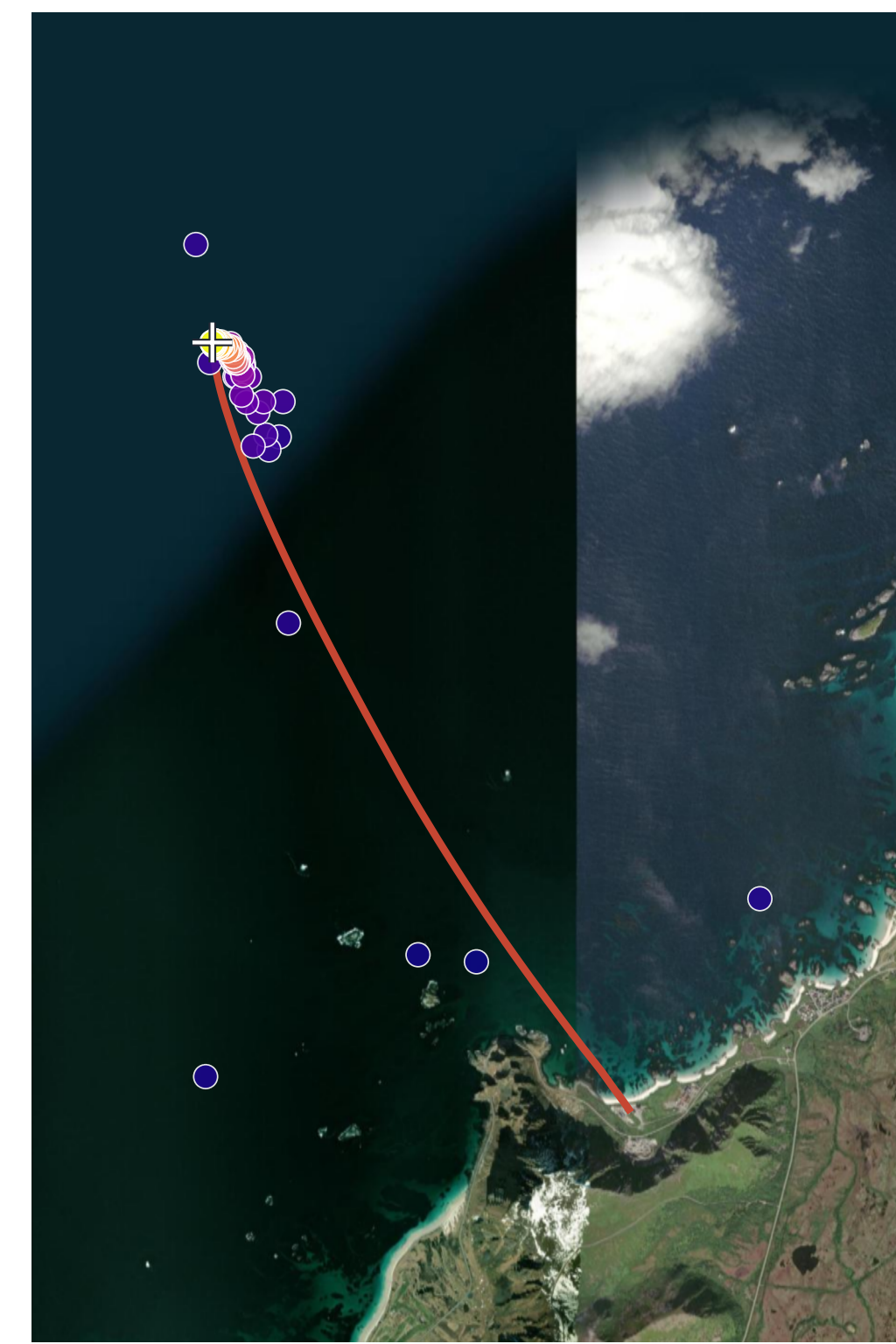
Simulation Performance

The Cartesian IPP architecture was evaluated on four simulated trajectories using 100 Monte Carlo wind realizations per trajectory. The rocket model was implemented with RocketPy and provides a controlled simulation reference without the hardware saturation seen in flight data.

The IPP achieved 45.3 m mean RMSE across the four trajectories. The plots show the median error and 5th–95th percentile spread.



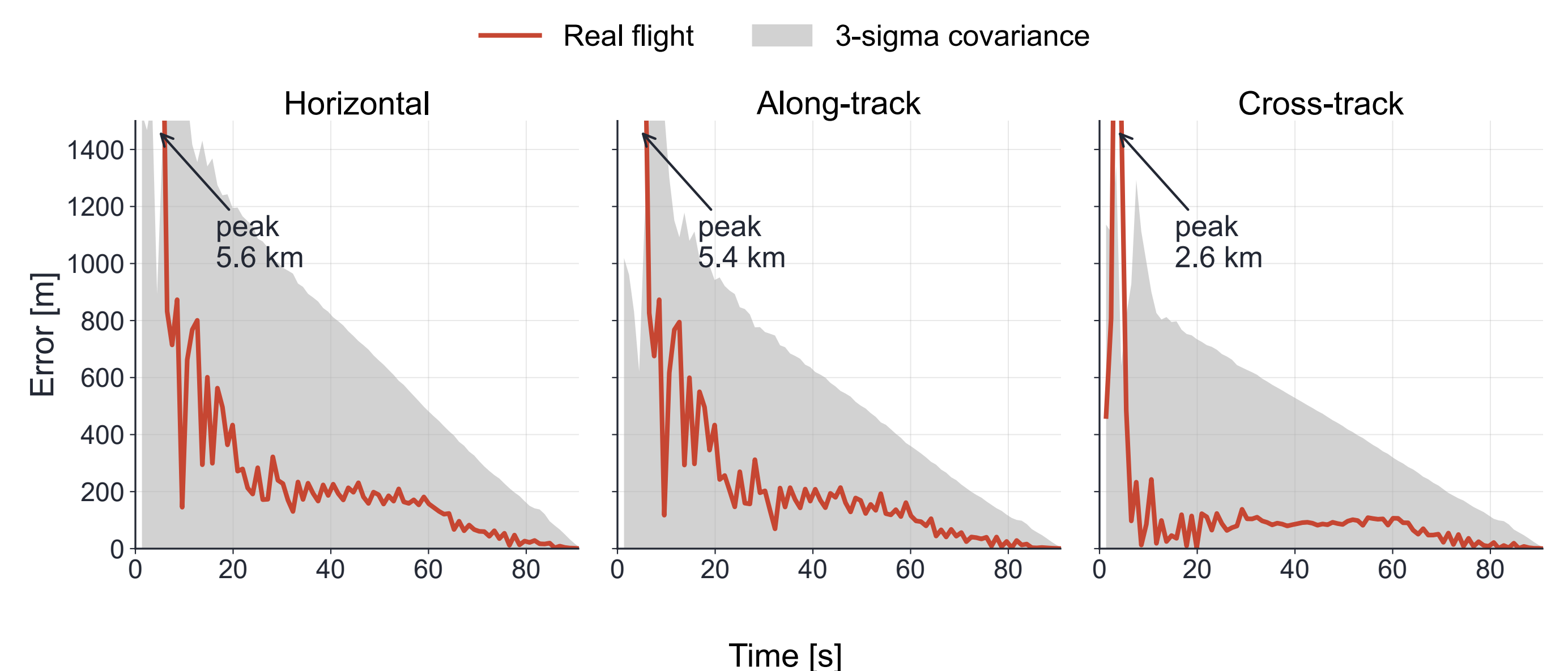
Andøya Test Flight



The IPP was also evaluated on an Andøya student rocket flight. The map shows the rocket track in red, IPP predictions coloured by flight time, and the actual impact marked by the crosshair.

Wind profiles came from a weather balloon launched 30–45 min before launch. Burn phase outliers are mainly explained by initial yaw uncertainty and attitude instability from gyroscope saturation under high spin. Thrust model errors may compound this. After burnout, the IPP error remains small through the ballistic phase.

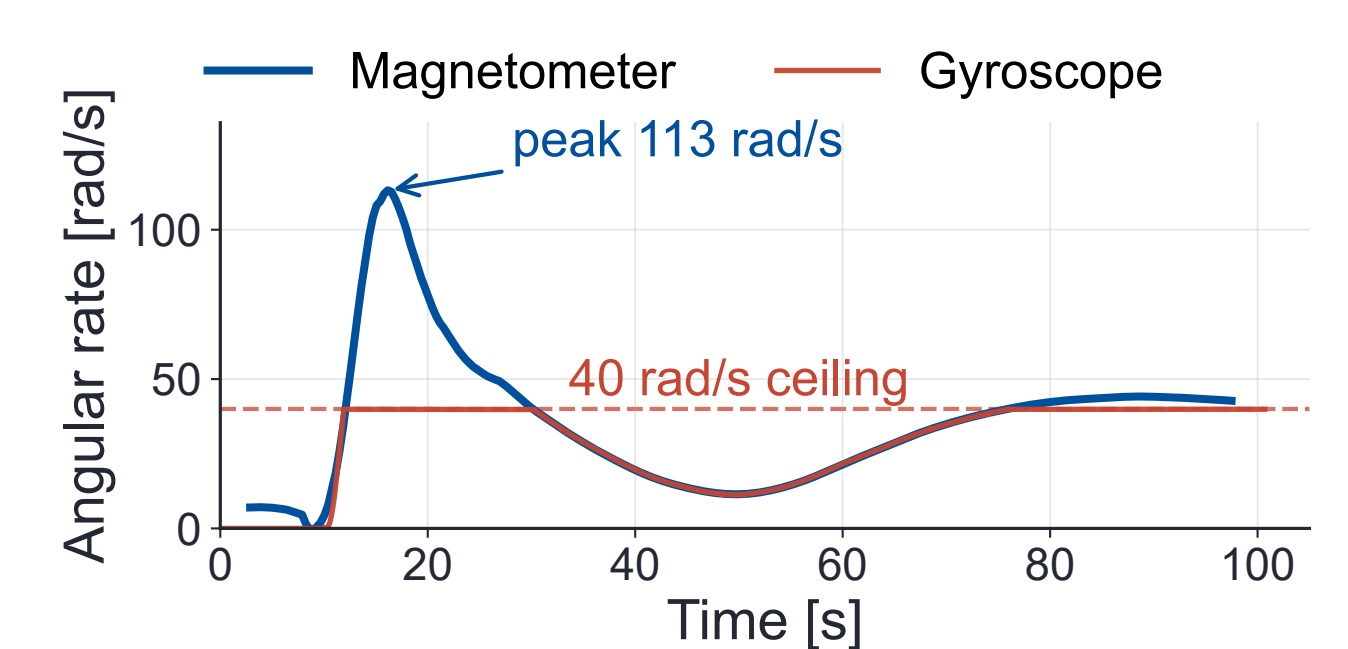
Real Flight IPP Error



The plot shows the real flight IPP error. Burn phase errors are dominated by sensor saturation and thrust uncertainty. After burnout, the error drops sharply.

Discussion

Simulation results show accurate IPP under controlled model conditions, but real flight exposes practical limits. The dominant issue was gyroscope saturation during burn, visible as the 40 rad/s ceiling in the spin-rate plot.



The high spin rate is a result of the rocket's spin stabilization, which active stabilization would avoid. Mounting the IMU closer to the spin axis or using a higher-range gyroscope would reduce saturation. Better wind estimation, drag identification, parameter estimation, and uncertainty quantification are key next steps.

References

- [1] K. Gryte et al. Recovery of Rocket Payloads and First Stages Using Unmanned Vehicles. IEEE Aerospace Conference, 2025.
- [2] A. Meta. Impact Point Prediction of High-Power Rockets and Sounding Rockets. M.Sc. Thesis, NTNU, 2025.
- [3] E. K. Jensen. Real-Time Impact Point Prediction for Autonomous Recovery of Suborbital Rockets. M.Sc. Thesis, NTNU, 2026.